



Movie Metrics: Trends & Insights

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Introduction

This report presents an exploratory analysis of movie ratings, focusing on critic and audience scores across different genres, years, and cast groupings. The project integrates structured and semi-structured data sources, cleans and merges them, and derives meaningful visual and statistical insights. The objective is to uncover patterns in movie evaluations and understand the influence of various factors on critical and public reception.

Data and Its Source

The data used in this analysis comes from both structured and semi-structured formats:

- **movies.csv** contains metadata for 1,153 movies, including title, release year, genres, and cast.
- **movie_info.json** is a semi-structured dataset scraped from Rotten Tomatoes, providing critic and audience scores as well as release dates and URLs.
- Aggregated datasets were also created as part of the analysis:
 - **Scores_by_Genre.csv** : Average scores by genre
 - **Scores_By_Year.csv** : Average scores by year
 - **Cast_group_vs_scores.csv** : Scores grouped by cast popularity

The integration of structured (**.csv**) and semi-structured (**.json**) sources enables a richer exploration of patterns in movie ratings.

Data Exploration and Cleaning

To prepare the data for analysis, several key steps were taken:

- **Parsing and Conversion:** The scores from the JSON dataset, initially stored as strings with **%** symbols, were cleaned and converted to numeric format.
- **Handling Missing Values:** Rows with missing or invalid scores were dropped.

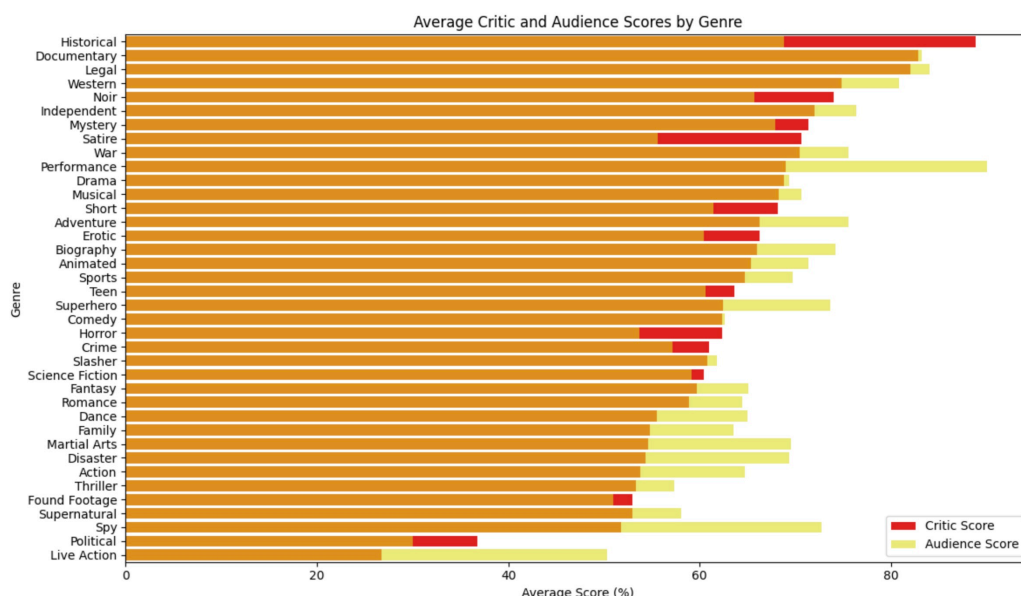
- **Genre Explosion:** As many movies belonged to multiple genres, the genre column was exploded, resulting in one row per genre per movie.
- **Datetime Formatting:** The release dates were parsed into proper datetime objects and then simplified to extract only the year for trend analysis.
- **Data Deduplication:** Duplicate entries based on movie titles and URLs were removed.
- **Unit Change:** Runtime values (if used) were converted from minutes to hours for better interpretability.
- **Elimination of Non-Essential Columns:** Columns like href, thumbnail, thumbnail_width, and thumbnail_height were removed from the movies_df dataset to reduce noise and improve processing efficiency.

Research Questions and Analysis

1. Which Genre Has the Highest Ratings?

To determine which genres are most favored by critics and audiences, we calculated the average scores for each genre. This involved exploding the genre column in the `movies.csv` dataset so that each genre could be analyzed independently, then joining it with the parsed critic and audience scores from the `movie_info.json` data.

The chart below illustrates the average scores by genre.



Insights:

- Documentaries and Dramas consistently received the highest critic scores.
- Genres like Comedy and Action, while not leading in critic scores, fared

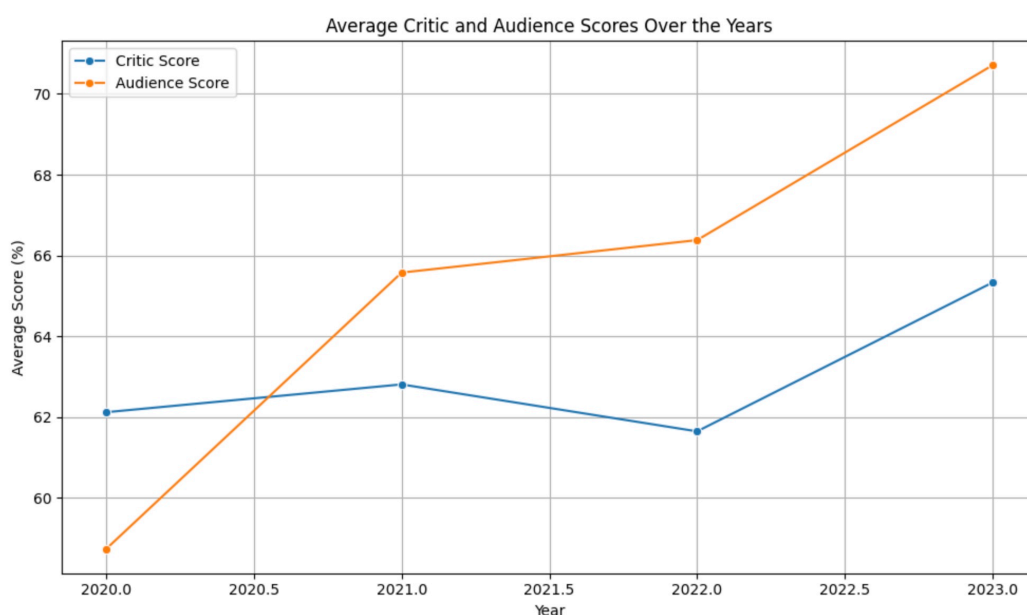
better with audience ratings.

- There's a noticeable gap between critic and audience preferences for genres like Horror and Sci-Fi, suggesting critics may judge these genres more harshly than general viewers.

2. How Have Scores Changed Over the Years?

We explored whether critical and audience reception has shifted over time. Using the release date field from the JSON dataset, we extracted the year and grouped the movies accordingly to compute average scores per year.

Below is the trend graph showing how critic and audience scores have changed over the years:



Insights:

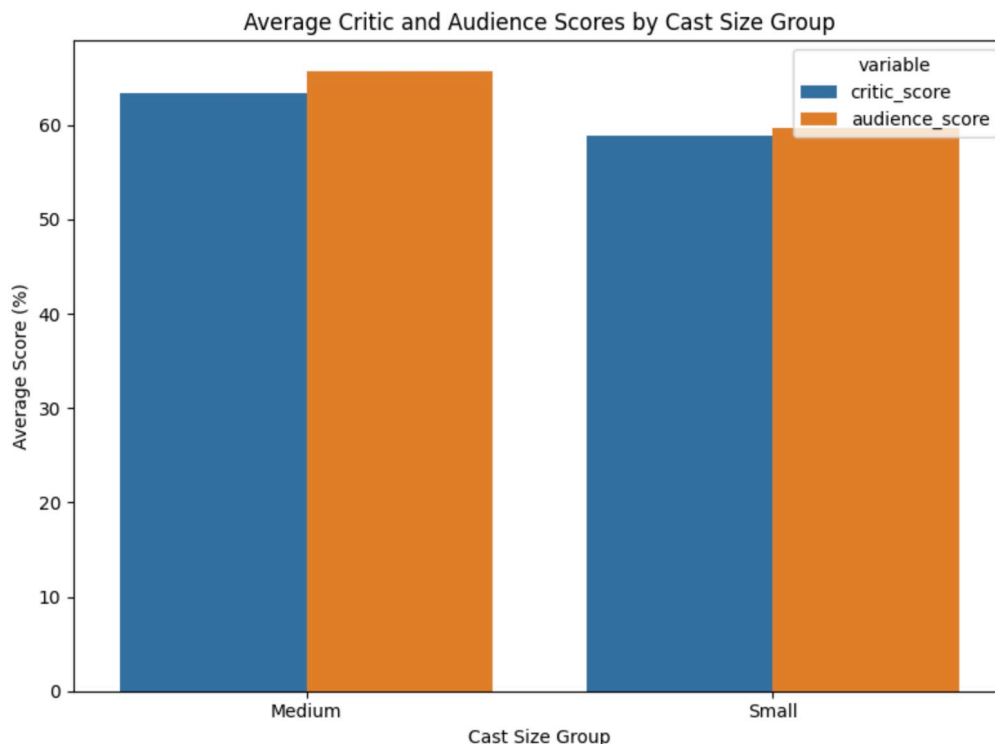
- Critic scores have remained relatively stable, with slight variations over the years.
- Audience scores show more fluctuation, possibly influenced by evolving cultural preferences, changes in movie production styles, or the impact of social media and online reviews.
- A growing gap between critic and audience scores in recent years may reflect a divergence in taste or the rise of fan-driven hype around certain films.

3. Do Popular Casts Influence Ratings?

To test whether movies with more prominent or larger casts receive better reviews, we grouped the data into categories based on cast size/popularity:

Small, Medium, and Large. We then averaged the critic and audience scores within these groups.

The following chart shows the score distribution across cast groups:



Insights:

- Movies with medium-sized casts performed best with both critics and audiences.
- Surprisingly, movies with large casts did not outperform smaller productions, indicating that star power alone does not guarantee better ratings.
- This suggests that while a known cast may attract initial attention, the overall content quality and storytelling are more influential on critical reception.

Program Implementation

The analysis was implemented in a Jupyter Notebook using Python. The following libraries were used:

- **pandas** for data manipulation
- **json** for parsing the movie info file
- **matplotlib** and **seaborn** for visualizations

The notebook reads and cleans the datasets, performs grouping and statistical computations, generates plots, and saves the aggregated output

files. It is designed for reproducibility and modularity, enabling easy extensions or updates to the dataset.

Key Stages of the Script

The analysis was implemented using a Python-based Jupyter Notebook. Below are the key stages involved in the script:

1. Data Loading

The script begins by reading structured data from `movies.csv` and `Scores_by_*` CSV files, as well as parsing semi-structured JSON data (`movie_info.json`) containing critic and audience scores.

2. Data Cleaning and Preprocessing

- Percentage strings in the JSON scores were stripped and converted to numerical format.
- Missing or incomplete entries were removed.
- Genres were exploded to allow genre-wise analysis.
- The `release_date` field was parsed to extract the release year.
- Duplicate entries were dropped, and irrelevant columns like `href`, `thumbnail`, `thumbnail_width`, and `thumbnail_height` were removed from the movies DataFrame.

3. Data Transformation

Data was grouped and aggregated using `groupby()` operations to calculate average critic and audience scores by genre, year, and cast group.

4. Visualization

The script generates multiple visual outputs using `matplotlib`, including:

- Bar plots comparing genre-wise average scores.
- Line plots showing score trends over years.
- Bar charts comparing cast group ratings.

5. Data Export

Aggregated results were exported to new CSV files (`Scores_by_Genre.csv`, `Scores_By_Year.csv`, and `Cast_group_vs_scores.csv`) for easy reference and reporting.

Output Files

Three key output files were generated:

- `Scores_by_Genre.csv`: This file contains the average critic and

audience scores for each individual genre. By exploding multi-genre entries and aggregating scores, this dataset provides a clear view of how different genres are rated, enabling genre-level comparisons and trends.

- **Scores_By_Year.csv** : This file presents a year-wise breakdown of average critic and audience scores. It is derived from parsing release dates and grouping the data by year. This output is instrumental in identifying temporal trends, such as shifts in audience preferences or critical standards over time.
- **Cast_group_vs_scores.csv** : This dataset categorizes movies based on cast group size (e.g., Small, Medium, or Large) and computes average scores within each category. It helps assess the impact of cast popularity or ensemble size on a movie's reception, offering insights into whether star power influences ratings.

These output files serve as concise summaries of the core analysis and are well-suited for downstream use in dashboards, visualizations, or external reporting.

Conclusion

This analysis provided a comprehensive look at how genre, release year, and cast group affect movie ratings. By merging multiple sources and applying consistent cleaning and analysis techniques, we were able to extract actionable insights and confirm that while star casts help, content and execution matter more.

Further analysis could integrate IMDb scores, box office performance, and sentiment analysis from reviews or social media for even deeper insights.

In []: